

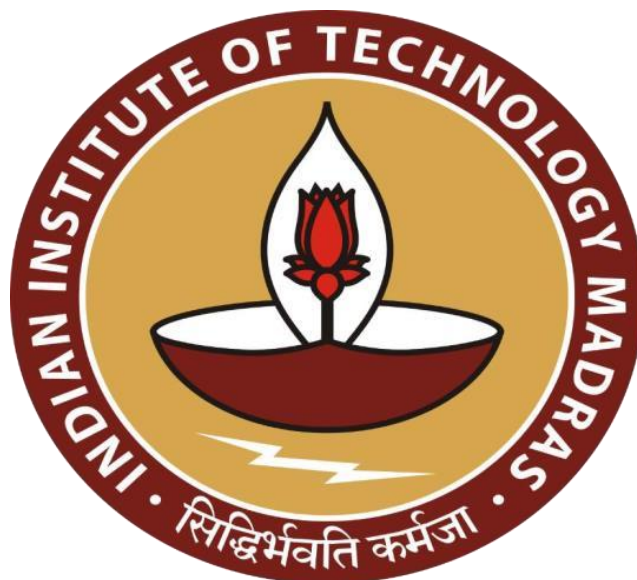
Data-Driven Inventory Planning for a Retail Pharmacy

Final-Term report for the BDM capstone Project

Submitted by

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1.Executive Summary

Rahul Medical Store is a local pharmacy situated in Sambalpur, Odisha. It offers a variety of prescription drugs, and health related products. From discussions with the owner, the following problems were identified:

- frequent expiry and wastage of medicines due to unstructured stock tracking,
- unrecorded or experience-based stock decisions without forecasting, and
- occasional non-prescription medicine sales that may affect store accountability.

The approach followed to do this capstone project is as follows:

Step 1: Define the business goals

Step 2: Decide how to measure and track them

Step 3: Collect and clean the data

Step 4: Analyze the data using Excel tools

Step 5: Visualize and interpret the results

From April to August 2025, information was gathered from the store's purchase and sales registers. Due to the shorter data duration and some mishandled expiry records, some of the results from the mid-term analysis were not entirely accurate; these have now been fixed in this final report.

The results have been fully visualized and interpreted in this final report. The results were displayed using charts like Column, Line, ABC Analysis, and Pareto. The findings indicate that a small number of expensive medications account for the majority of sales, while a number of slow-moving items raise the risk of expiration.

In order to reduce waste, enhance reorder planning, and uphold accountability in the sale of non-prescription medications, Rahul Medical Store must implement a systematic, data-driven inventory model.

2.Detailed Explanation of Analysis Process

1.Introduction

- The purpose of this analysis is to improve the inventory management system of Rahul Medical Store by using real transaction data. The dataset covers a period of five months, from April to August 2025, and includes the daily sales and purchase records of medicines and health products at Rahul Medical Store in Sambalpur, Odisha.
- The main aim of the analysis was to identify sales patterns, track stock movement, and detect expiry risks using simple and practical tools that can be adopted by the store owner without technical expertise.

- Two major types of analysis were conducted:
 - **Descriptive Analysis** – to understand what has happened, such as total sales, purchases, and product performance across months.
 - **Diagnostic Analysis** – to explore why certain problems occurred, such as overstocking, expiry risks, or low turnover in specific items.
- All calculations, summaries, and visualizations were performed in “Microsoft Excel” using pivot tables, filters, formulas, and charts. These methods were chosen because they are transparent, easy to replicate, and suitable for everyday business use.
- The results of this analysis provide a basic foundation for decision-making, helping the store identify profitable products, reduce expiry-related losses, and plan future purchases more efficiently and productively.

2.Methodology and Tools Used

To guarantee the accuracy and applicability of the findings, the entire project adhered to a methodical and structured approach. From data collection to actionable insights, the workflow followed a methodical process. First, the store's business challenges were identified, and then the right Excel-based analytical tools were chosen to address them. Data was digitized into Excel for simpler manipulation after being taken out of the store owner's manually kept purchase and sales registers.

Microsoft Excel was selected as the primary analytical platform because of its wide accessibility, ease of use, and suitability for small business data analysis. Key Excel tools such as Pivot Tables, Pivot Charts, Conditional Formatting, and Formulas (like IF, SUMIFS, and DATE) were utilized extensively. These tools helped in summarizing large transactional data into meaningful patterns that could be easily interpreted by the store owner. Basic statistical functions such as averages, cumulative percentages, and proportions were also used to understand overall sales and purchase trends.

The analytical design emphasized simplicity and reproducibility. This means that the same method can be reused by the store every month with minimal technical effort. The dataset was kept modular, allowing separate sheets for sales and purchase data, making future extensions—such as forecasting or automation—easier to implement.

3.Objective

- The main objective of this project is to improve the inventory management process of Rahul Medical Store using data collected from its sales and purchase registers.
- The specific objectives are:
 1. To evaluate stock movement and expiry trends to reduce wastage.
 2. To analyse the impact of non-prescription medicine sales on store accountability.
 3. To propose a data-driven model to optimize stock management using Excel-based tools.

4.Pre-processing of the Data

The raw data was collected from the sales and purchase registers of Rahul Medical Store and manually entered into Excel. It contained information such as Date, Product Name, Batch Number, Expiry Date, Quantity, MRP, GST, and Total Amount.

The dataset covered transactions for five months (April–August 2025) and included both sales and purchase records.

Before conducting any analysis, the data was examined and cleansed to guarantee accuracy. Blank or duplicate rows were eliminated, and formatting mistakes like inconsistent date or number formats were fixed. While text fields like Product Name and Batch Number were examined for errors or missing entries, the Quantity, MRP, and Amount columns were confirmed to contain valid numerical values.

There were a few records with missing dates, such as those for specific items like thermometers, but these were few and had little bearing on the analysis as a whole. Whether these thermometers were mercury-based or digital was not made clear by the owner. This distinction is significant because digital thermometers may need to be replaced on a regular basis because of battery or sensor problems, whereas mercury thermometers typically have no expiration date. As a result, these documents were kept but excluded from the assessment of expiry risk.

The data was discovered to be clean, consistent, and prepared for analysis following the cleaning procedure. The final dataset was saved as a master Excel file and used for descriptive and diagnostic analyses such as sales trends, ABC categorization, and expiry risk evaluation

RAHUL MEDICAL STORE,Sambalpur,Odisha											
Purchase Register											
Period:- April 25 - August25											
Date	Customer/Supplier	Product Name	Batch No	Expiry Date	Quantity	Rate (₹)	Amount (₹)	GST (%)	T Amount	Total (₹)	Expiry Status
2025-04-01	Popular Medical Store	Allegra 120mg	104	Jul-26	10	211.78	1890.89	12	226.91	2117.80	Safe
2025-04-01	Popular Medical Store	Multivitamin Syrup	892	Dec-25	10	131.61	1175.09	12	141.01	1316.10	At Risk
2025-04-01	Taj Pharmsutical	Clotrimazole Cream	725	May-26	25	187.55	4186.38	12	502.37	4688.75	Safe
2025-04-01	Popular Medical Store	Metformin 500mg	115	Apr-26	10	20.68	184.64	12	22.16	206.80	Safe
2025-04-01	Utkal Pharma	Ibuprofen 400mg	395	May-26	10	7.63	68.13	12	8.18	76.30	Safe
2025-04-01	Utkal Pharma	Combiflam	442	Jun-26	50	44.57	1989.73	12	238.77	2228.50	Safe
2025-04-02	Popular Medical Store	Zincovit	734	Mar-26	10	82	732.14	12	87.86	820.00	Safe
2025-04-02	Siba Medical Store	Amlodipine 5mg	600	Jan-26	10	71.31	636.70	12	76.40	713.10	At Risk
2025-04-03	Odisha Chemist & drugist	Volini Gel	994	Jan-26	25	218.36	4874.11	12	584.89	5459.00	At Risk

Table1: Sample of Cleaned Purchase Register Dataset after Pre-processing

5.Descriptive Analysis

1.Monthly Sales & Purchase Trend

The cleaned dataset was analyzed to understand Rahul Medical Store's purchase and sales trends from April to August 2025. Descriptive analysis's primary objective is to identify what has happened, such as total sales, purchase volumes, and significant products that generate revenue. All analyses were conducted using Microsoft Excel, which included pivot tables and charts for better visualization and interpretation.

The condensed sales data (April–August 2025) was used to create a pivot chart. With May having the highest total and April having the lowest, the chart clearly illustrates monthly sales fluctuations. Following the initial procurement phase, June, July, and August showed consistent sales momentum with stable performance.

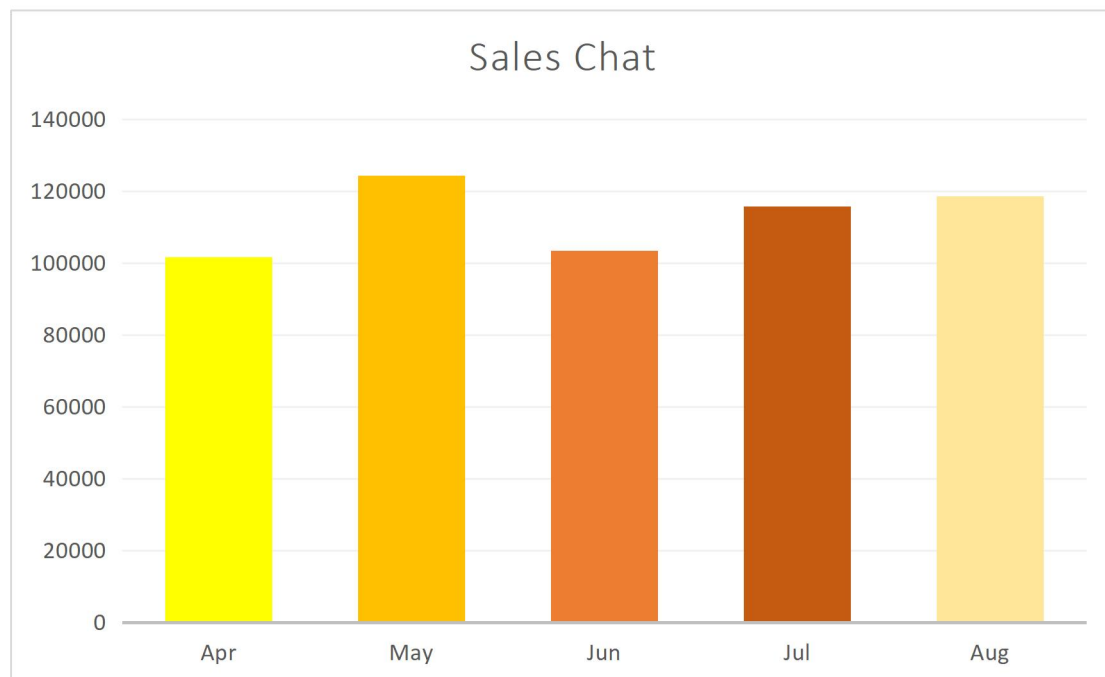


Figure 1: Monthly Sales Trend of Rahul Medical Store (April–August 2025)

Similarly, the Purchase Chart in Figure 3 represents the month-wise total purchase values for the same period. Purchases were also highest in April and lowest in June

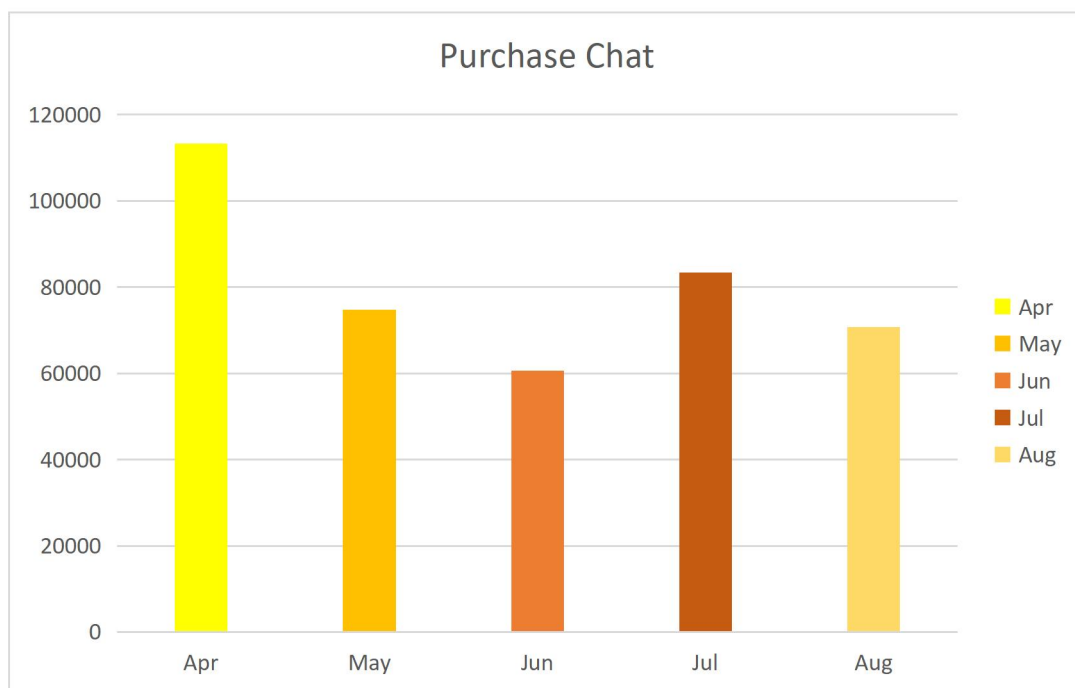


Figure 2: Monthly Purchase Trend of Rahul Medical Store (April–August 2025)

Sales peaked in May and declined in April, indicating that demand rose after the first month, possibly as people restocked on common drugs and medical supplies. After May, sales in June, July, and August varied very little, suggesting consistent demand.

Purchases peaked in April, the store's main inventory procurement period at the start of the fiscal year, and then sharply declined in June, the month with the lowest restocking rates. A modest increase in July and a moderate level in August indicate strategic restocking based on product movement.. Overall, the study demonstrates that Rahul Medical Store maintained

balanced procurement in the following months to prevent overstocking and expiration risks by heavily replenishing stock in April ahead of May's peak sales.

2.Combined Sales and Purchase Trend

Month	Sales (₹)	Purchase (₹)
Apr-25	101604.5	113276.8
May-25	124272.3	74483.69
Jun-25	103507.5	60699.41
Jul-25	121271.1	82787.51
Aug-25	118691.1	71409.9

Table2: Monthly Sales and Purchase Values

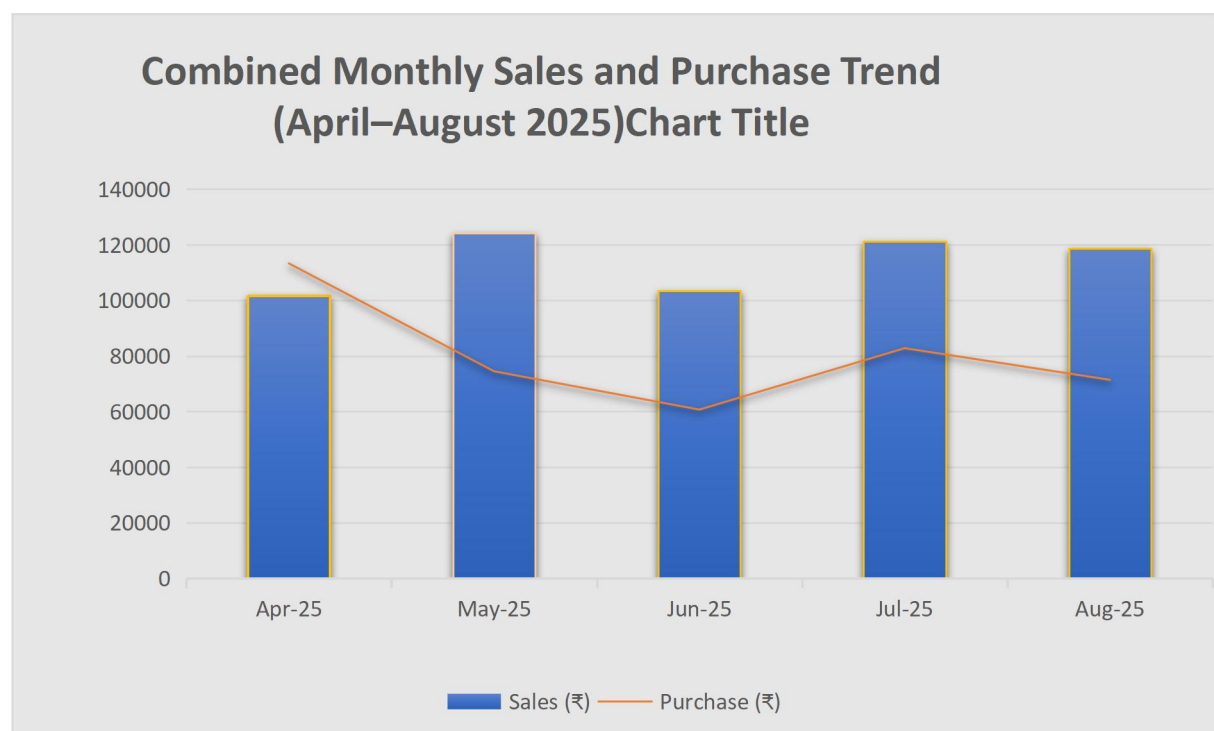


Figure 3: Combined Monthly Sales and Purchase Trend (April–August 2025)

Rahul Medical Store's combined sales and purchase trend from April to August 2025 demonstrates the connection between sales and procurement performance. Sales peaked in May, demonstrating effective use of this inventory, while purchases peaked in April, reflecting the procurement of bulk stock at the start of the fiscal period. Sales and purchases showed a balanced pattern from June to August, indicating steady demand and managed replenishment. All things considered, the two trends' synchronization shows good inventory control and planning.

Month	Total Sales (₹)	Total Purchases (₹)	Observation
25-Apr	1,01,604.5	1,13,276.8	Initial stock buildup
25-May	1,24,272.3	74,483.70	Peak sales month
25-Jun	1,03,507.5	60,699.40	Lowest purchase month
25-Jul	1,21,271.1	82,787.50	Demand stable
25-Aug	1,18,691.1	71,409.90	Moderate restocking

Table 3: Summary of Monthly Sales and Purchases (April–August 2025)

Overall, Rahul Medical Store's sales and purchase dynamics from April to August 2025 were clearly understood thanks to the descriptive analysis. The findings show that the store makes wise purchasing decisions and maintains a reasonably steady sales trend, meeting customer demand without going overboard. The analysis also emphasizes the store's proactive restocking strategy, which supported the month of May's peak sales with pre-purchased items. These results serve as the basis for additional research on expiry risks and inventory efficiency, which are examined in the diagnostic analysis that follows.

6.Diagnostic Analysis

Although the descriptive analysis was helpful in understanding Rahul Medical Store's overall sales and purchase behavior, it falls short in explaining why some patterns—like the high procurement in April, the sales spike in May, or the variation in stock movement across products—occurred. A diagnostic analysis was carried out in order to learn more about these patterns. To improve control over stock management and lower financial losses, this phase focuses on identifying the major contributors to total expenditure, evaluating expiry-related risks, and assessing inventory efficiency through ABC classification.

1.ABC Classification

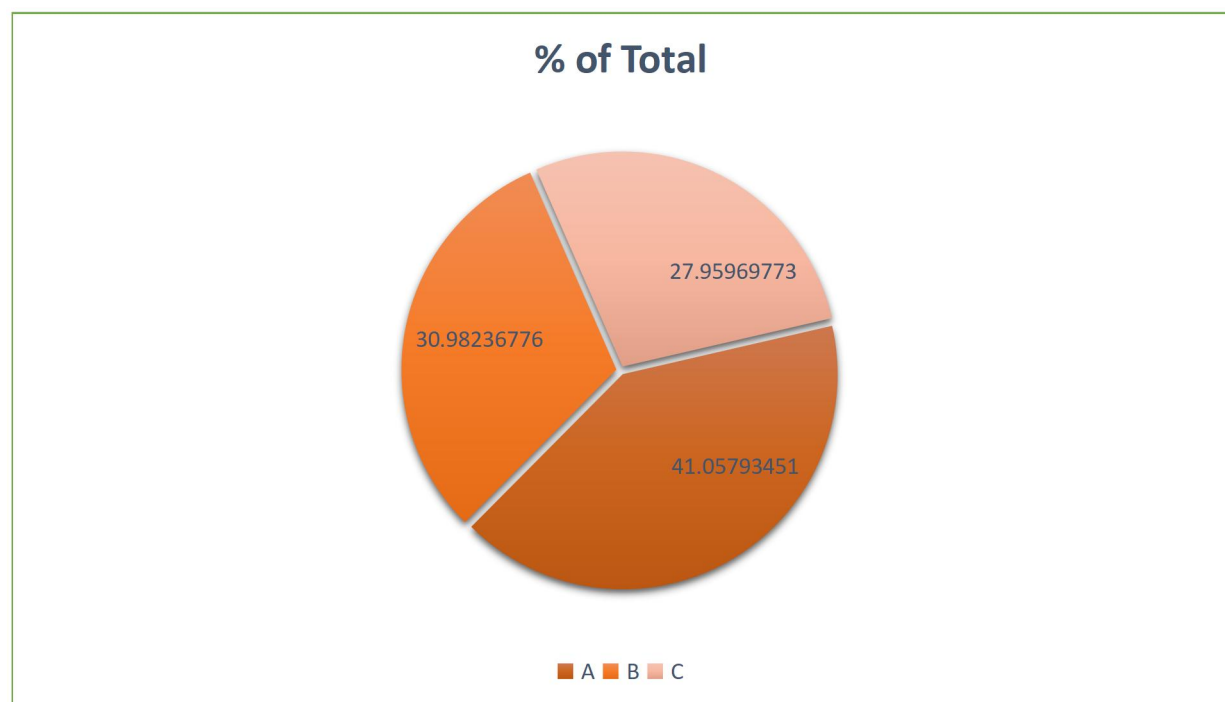


Figure 4: ABC Classification of Medicines at Rahul Medical Store (April–August 2025)

The ABC analysis in Figure 4 categorizes medicines based on their total purchase value from April to August 2025. As shown, Category A items account for approximately 41% of all medicines, representing the high-value products that contribute the most to total purchase

expenditure. Category B covers around 31%, indicating moderately priced or medium-movement items, while Category C represents the remaining 28%, consisting of low-value or slow-moving medicines.

This classification highlights that Rahul Medical Store should closely monitor A-category medicines to ensure optimal stock levels and avoid overinvestment, while B and C items can be managed with routine restocking. Overall, the analysis supports efficient inventory control and prioritization of critical stock items.

Category	Count	% of Total
A	163	41.05793451
B	123	30.98236776
C	111	27.95969773
	397	

Table 4: The quantitative summary of ABC Classification of Medicines (April–August 2025)

To interpret these results meaningfully, the categories were further analyzed based on their inventory control importance, as shown below.

Category	Description	% of Total Value	Inventory Control Priority
A	High-value and fast-moving medicines	41%	Very High
B	Moderately priced and medium-movement items	31%	Medium
C	Low-value, slow-moving or infrequent sales items	28%	Low

Table 5: Inventory Control Interpretation of ABC Categories

2.Expiry Risk Analysis

Status	Count	% of Total
Safe	304	76.5743073
At Risk	93	23.4256927
Total	397	

Table 6: *Summary of Expiry Status Classification of Medicines*

To assess the shelf-life status of the medications in the purchase register, an expiry risk analysis was carried out in addition to value-based classification. Using a date-based condition, the "Expiry Status" column automatically categorized each medication as either Safe or At Risk. In this study, medications that expire between August 2025 and January 2026 were classified as "At Risk," and medications that expire after January 2026 were classified as "Safe."

Only a small percentage of the inventory is approaching expiration, according to the analysis, which suggests efficient stock rotation and prompt procurement. The majority of products are classified as safe, guaranteeing a long enough shelf life for ongoing use and sale. To reduce possible losses from expiration, the At-Risk medications—mostly slow-moving or short-shelf-life items—should be regularly checked, marked down, or sent back to suppliers.

The Expiry Risk Chart (Figure 5) visualizes this distribution, highlighting that the majority of the store's stock remains stable, with a minor percentage requiring urgent attention to prevent wastage.

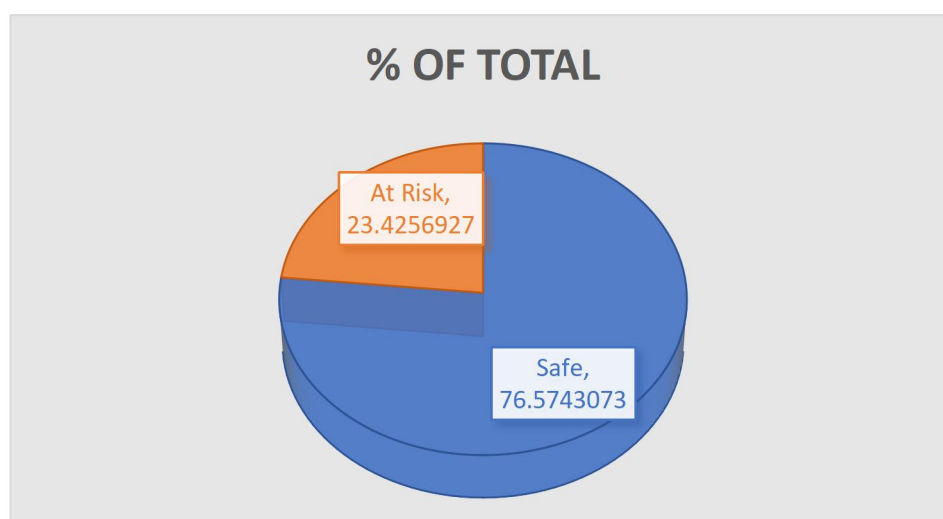


Figure 5: Expiry Status Distribution of Medicines (Safe vs. At-Risk)

Rank	Product Name	Supplier	Expiry Date	Total Value (₹)	Remarks
1	Volini Gel	Odisha Chemist & Drugist	26-Jan	₹27,295.00	High-value stock nearing expiry; multiple batches
2	Multivitamin Syrup	Popular Medical Store	25-Dec	₹20,531.16	Moderate-value item; slow movement risk
3	Insulin 30/70	Popular Medical Store	26-Jan	₹19,995.50	Essential medicine with moderate expiry exposure

Table 7: Top 3 High-Value Medicines Approaching Expiry (April–August 2025)

The above table identifies the top three medicines contributing the most to the store's expiry risk value. Volini Gel, sourced from Odisha Chemist & Drugist, represents the highest potential loss of ₹27,295 due to multiple batches nearing expiry in January 2026. Multivitamin Syrup and Insulin 30/70, supplied by Popular Medical Store, add another ₹20,531 and ₹19,995 respectively, highlighting significant stock tied up in near-expiry items. These products together constitute a major portion of the at-risk inventory. To mitigate this risk, Rahul Medical Store should prioritize sales of these medicines through discounts, promotional offers, or supplier returns to prevent financial losses.

Observations:

- Majority of near-expiry medicines belong to the high-value category.
- Overstocking without considering expiry timelines causes unnecessary capital blockage.
- Regular expiry audits can help identify and address such risks early.

Overall, the expiry analysis provides valuable insight into how capital gets locked in due to unsold or soon-to-expire products. Addressing these issues early can help the store maintain healthier inventory cycles. The next section, Stock Age and Movement Analysis, further explores how effectively the store's inventory is rotating and identifies areas where product movement can be improved.

3. Stock Age and Movement Analysis

Products	Average of Stock Age	Sum of Total (₹)
Disprin	7.0	577.28
Cetirizine 10mg	7.0	892.5
Calpol 500mg	7.0	932.17
Ibuprofen 400mg	7.0	877.45
Tetanus Toxoid	6.9	1626
Metformin 500mg	6.9	1964.6
Paracetamol 500mg	6.9	1934.4
Avil 25mg	6.7	1442
Levocetirizine 5mg	6.6	1834.35
Glibenclamide 5mg	6.6	2269.05

Table 8

Summary of Average Stock Age and Stock Value of Medicines (as of November 2025)

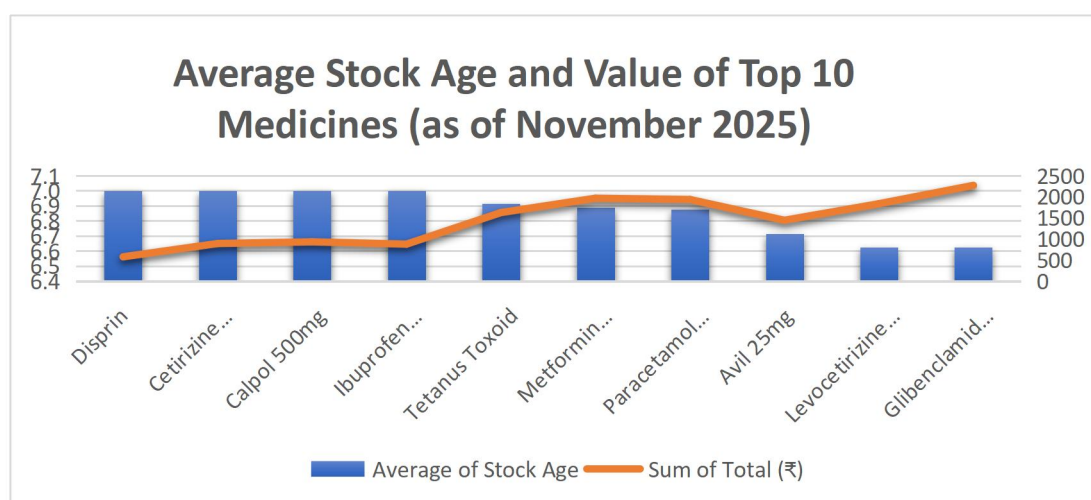


Figure 6: Average Stock Age and Value of Top 10 Medicines (as of November 2025)

The top ten medications with the highest average stock age are shown in the above chart. Disprin, 500 mg of Calpol, 400 mg of Ibuprofen, and 10 mg of Cetirizine have been in stock for almost seven months, which suggests poor movement and a greater chance of stagnation. Despite having comparatively low total values, these products' extended storage can affect how much shelf space is used. Conversely, medications with shorter stock ages—like Avil 25 mg and Paracetamol 500 mg—show faster rotation and steady sales.

In order to maintain the best possible balance between demand and stock availability, Rahul

Medical Store can use the combined analysis of stock age and value to identify slow-moving medications, schedule promotional sales, and modify reorder frequencies.

After analyzing the store's sales, purchases, product categories, and expiry risks through diagnostic analysis, it became clear that Rahul Medical Store has a well-functioning inventory but faces challenges in areas like overstocking, slow-moving items, and medicines nearing expiry.

While the diagnostic phase helped in understanding why these issues occur, the next step is to determine what actions can be taken to address them effectively.

The following Prescriptive Analysis section provides practical, data-driven recommendations aimed at improving inventory turnover, minimizing expiry-related losses, and ensuring more efficient purchase planning.

3. Prescriptive Analysis

After conducting descriptive and diagnostic analyses, the next step is to identify what actions Rahul Medical Store should take to improve operations, reduce wastage, and maintain consistent profitability.

The prescriptive analysis translates insights into practical, data-backed recommendations that the store can easily implement.

Smarter Purchase Planning Based on Sales Trends

- The analysis of monthly sales and purchases showed that purchases were highest in April, while sales peaked in May.
- This pattern suggests that the store stocked up before customer demand increased — a good practice, but it can lead to overstocking if not monitored.
- Rahul Medical Store should plan purchases based on real sales data from the previous month.
- Setting a reorder level (for example, restock when only 20% of an item remains) will ensure products are available without locking up extra money in unused stock.
- This will help maintain balance between demand and inventory while preventing excess procurement.

Prioritize A-Category Medicines (From ABC Analysis)

- The ABC classification showed that A-category medicines contribute over 40% of total purchase value, even though they make up only about a third of total items.
- These are the most valuable and fastest-selling products — often essential medicines that customers buy regularly.
- The store should:
 - Monitor A-category items daily to avoid stockouts.
 - Negotiate bulk purchase discounts with suppliers for high-value drugs.

- Avoid over-purchasing them, as large quantities increase the risk of expiry.
- Regular review of Category A medicines will help ensure steady profit margins and better stock turnover.

Act Quickly on At-Risk Medicines (From Expiry Risk Analysis)

- Around 23% of the total stock was found to be in the “At-Risk” category — meaning these medicines will expire within 5–6 months.
- These are a potential source of financial loss if not addressed in time.
- To manage them efficiently:
 - Identify and tag “At-Risk” items every month.
 - Promote them through small discounts or combo offers to sell them faster.
 - Return or exchange near-expiry stock with suppliers wherever possible.
 - Keep such medicines in a separate shelf or labeled section for easy tracking.
- By doing this, Rahul Medical Store can minimize wastage and recover maximum value from soon-to-expire stock.

Reduce Stock Holding Time (From Stock Age Analysis)

- The Stock Age analysis revealed that certain medicines like Disprin, Calpol 500mg, Ibuprofen 400mg, and Cetirizine 10mg have been in stock for nearly seven months.
- This indicates low movement or excess stock, which can lead to expiry and lost shelf space.
- The store should:
 - Implement a First-Expire-First-Out (FEFO) approach so older batches are sold before new ones.
 - Track average stock age monthly to identify items that aren’t selling well.
 - Offer small discounts or bundle packs for slow-moving products.
- This will improve cash flow, reduce idle stock, and free up space for high-demand medicines.

Digitize Inventory Tracking for Better Control

- Currently, stock management and expiry tracking are done manually in Excel — which can lead to delays or errors.
- Rahul Medical Store can benefit greatly by digitizing its inventory system using tools it already knows (like Excel or Google Sheets).
- Suggested steps:
 - Maintain a monthly dashboard showing sales, purchases, expiry risk, and top-performing items.
 - Use Conditional Formatting in Excel to automatically highlight “At-Risk” or low-stock medicines.
 - Store data online for easy access and backup.

- Over time, this will make the system more efficient, transparent, and easier to scale.

Strengthen Control Over Non-Prescription Medicine Sales

- During data collection, it was observed that certain medicines were sold without prescriptions.
- While these are commonly used drugs (like pain relievers or cold medicines), frequent unrecorded sales can affect the store's accountability and regulatory compliance.
- Rahul Medical Store should take the following steps to manage non-prescription sales more transparently:
 - Maintain a separate record for all non-prescription medicines in Excel or a small register.
 - Tag such items clearly in the inventory file (e.g., "OTC" for over-the-counter).
 - Periodically review sales trends of these products to ensure they don't exceed reasonable limits.
 - Provide customers with clear guidance and restrict bulk sales of sensitive items.
- This will improve accountability, ensure ethical sales practices, and help the store maintain a positive compliance record with local health authorities.

4.Future Scope

This project still has a lot of room to grow. A more comprehensive understanding of seasonal trends, demand patterns, and long-term performance would be possible with data analysis conducted over a longer time frame, such as a full year. Rahul Medical Store would be able to better plan purchases and reduce shortages or overstocking as a result.

The store may eventually use interactive dashboards or simple forecasting tools to see real-time sales, stock levels, and expiration dates. Inventory tracking can be made easier and more efficient with the use of programs like Google Sheets, Power BI, or basic mobile apps.

To aid in data-driven decision-making, it would also be beneficial to incorporate sales prediction models, supplier performance analysis, and automated expiry alerts.

Rahul Medical Store can become a data-smart, well-run, and sustainable company by implementing these minor technology changes gradually. This will serve as a model for other nearby pharmacies..

5.Conclusion

Rahul Medical Store's inventory management practices were clearly illustrated by this project, which also pointed out areas in need of development. I gained an understanding of how sales, purchases, and expiration dates affect the store's performance by examining data from April

to August 2025.

Despite stable sales, issues like slow-moving inventory, overbuying, and medications that are about to expire still result in waste and capital blockages. I determined which products generate the majority of sales and which require more attention by using Excel-based analyses like ABC classification, expiry risk tracking, and stock age analysis.

The recommendations for the store to utilize the First-Expire-First-Out (FEFO) method, keep digital records, and purchase inventory based on the data from sales trends involve strategies to minimize waste in products and improve efficiency, which is simpler to do based on sales trends and also make decisions regarding finances. The proposal to distinguish sales of non-prescription items independently increases transparency and accountability in the pharmacy's operations.

In summary, this project exemplifies how small community pharmacies can engage in significant improvement based on simple data analysis. This experience highlighted that data-influenced thinking is not only for large companies, with the right mindset even simple tools like Excel can lead to improved reasoning in the decision making of daily business functions into smarter systems that are more efficient and sustainable.